

2. Components of an Experiment II

Review: What form of Causal Reasoning is being applied? What are its strengths and weaknesses?

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40 Cal students who reported exercising 3+ times/week reported lower stress levels than 40 who reported exercising 0-2 times per week, therefore,

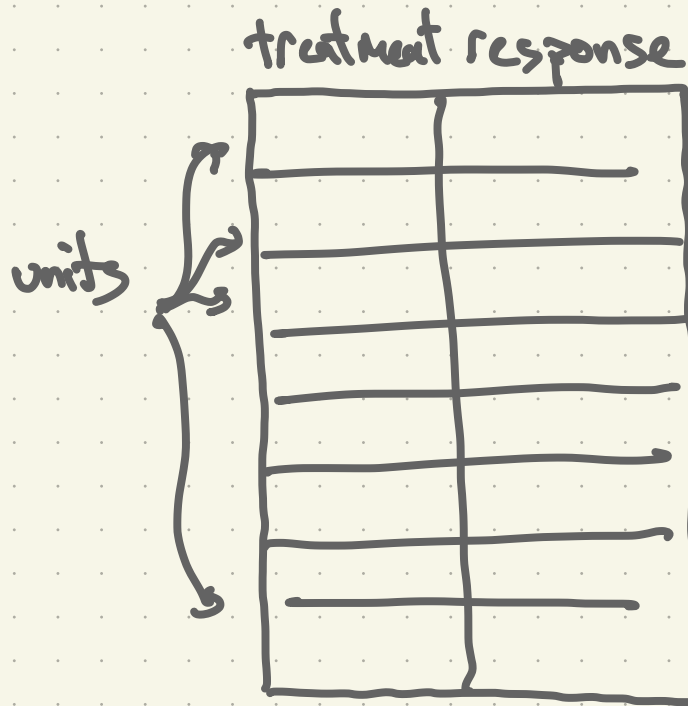
for those 80 students, regular exercise causes decreased stress levels on average.

Components of an Experiment

An experimental design is characterized by:

- ① the nature of the treatments
- ② choice of experimental units
- ③ manner of assigning units to treatments.
- ④ the nature of the response.

Protocol



Protocol

- ① ~~~
- ② ~~~
- ③ ~~~

① The Nature of the Treatments

We chose an arbitrary number between 0 and 100.

The number selected and assigned to you is $X = 11$.

1. Do you think the percentage of countries, among all those in the United Nations, that are in Africa is **higher** or **lower** than X ? (circle one)

Higher

Lower

2. Give your best estimate of the *percentage* of countries, among all those in the United Nations, that are in Africa: _____

We chose an arbitrary number between 0 and 100.

The number selected and assigned to you is $X = 73$.

1. Do you think the percentage of countries, among all those in the United Nations, that are in Africa is **higher** or **lower** than X ? (circle one)

Higher

Lower

2. Give your best estimate of the *percentage* of countries, among all those in the United Nations, that are in Africa: _____

Q: How could you modify the treatments to decrease/increase the effect?

② Choice of Experimental units

② Choice of Experimental units

An
Alternate
Design

Imagine we had conducted the anchoring experiment in lab instead of lecture. Lab

1 receives $X=11$ and Lab 2 receives $X=73$.

What is the experimental unit?

treatment response

11	
11	
11	
11	
73	
73	
33	
73	

③ Manner of Assigning Units

What we did: Shuffle the handouts + pass them out haphazardly.

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How can this method be improved?

•

④ The nature of the response

→ "The % of UN nations that are in Africa".

Types of Variability in the Response

①

②

③